Single Use Negative pRessure dressing for Reduction in Surgical site infection following Emergency laparotomy: The SUNRRISE Trial

Trial Registration: ISRCTN 17599457

Statistical Analysis Plan

SAP Version Number	Protocol Version Number
1.0	4.0

Name of Author:	Samir Mehta	Role:	Trial Statistician	Affiliation:	BCTU University of Birmingham
Signature of Author:		Date:			
Name of Chief Investigator:	Richard Wilkin	Role:	Chief Investigator	Affiliation:	
Signature of Chief Investigator:		Date:			
Name of Co-Chief Investigator:	Hamish Clouston	Role:	Co-Chief Investigator	Affiliation:	
Signature of Chief Investigator:		Date:			
Name of reviewing statistician:	Rebecca Woolley	Role:	Reviewer	Affiliation:	BCTU University of Birmingham
Signature of reviewing statistician		Date:			
This Statistical Analysis Plan has been approved by:					
Name of Approver:	Natalie Rowland	Role:	Senior Statistician	Affiliation:	BCTU University of Birmingham
Signature of Approver:		Date:			

Statistical Analysis Plan (SAP) Amendments

<table>
<tr><th>SAP version number</th><th>SAP section number</th><th>Description of and reason for change</th><th>Timing of change with respect to interim analysis/ final analysis/ database lock</th><th colspan="2">Blind Reviewer</th></tr>
<tr><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td>Name:</td><td></td></tr>
<tr><td>Signature:</td><td></td></tr>
<tr><td>Date:</td><td></td></tr>
<tr><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td>Name:</td><td></td></tr>
<tr><td>Signature:</td><td></td></tr>
<tr><td>Date:</td><td></td></tr>
<tr><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td rowspan="3"></td><td>Name:</td><td></td></tr>
<tr><td>Signature:</td><td></td></tr>
<tr><td>Date:</td><td></td></tr>
</table>

Abbreviations & Definitions	
Abbreviation / Acronym	**Meaning**
BCTU	Birmingham Clinical Trials Unit
CDC	Centers for Disease Control and Prevention
CI	Confidence Interval
CONSORT	Consolidated Standards of Reporting Trials
CRF	Case Report Form
DMC	Data Monitoring Committee
ISRCTN	International Standard Randomised Controlled Trial Number
ITT	Intention to Treat
MCS	Mental component score (of SF-12)
NPWT	Negative Pressure Wound Therapy
PCS	Physical component score (of SF-12)
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SF-12	Short Form-12
SSI	Surgical Site Infection
SUNPD	Single-use Negative Pressure Dressing
URSAE	Unexpected and Related Serious Adverse Events
TSC	Trial Steering Committee
Term	**Definition**
International Standard Randomised Controlled Trial Number	A clinical trial registry
Protocol	Document that details the rationale, objectives, design, methodology and statistical considerations of the study
Randomisation	The process of assigning trial subjects to intervention or control groups using an element of chance to determine the assignments in order to reduce bias.
Statistical Analysis Plan	Pre-specified statistical methodology documented for the trial, either in the protocol or in a separate document.

TABLE OF CONTENTS

1. Introduction

This document is the Statistical Analysis Plan (SAP) for the SUNRRISE trial, and should be read in conjunction with the current trial protocol. This SAP details the proposed analyses and presentation of the data for the main paper(s) reporting the results for the SUNRRISE trial.

The results reported in these papers will follow the strategy set out here. Subsequent analyses of a more exploratory nature will not be bound by this strategy, though they are expected to follow the broad principles laid down here. The principles are not intended to curtail exploratory analysis (e.g. to decide cut-points for categorisation of continuous variables), nor to prohibit accepted practices (e.g. transformation of data prior to analysis), but they are intended to establish rules that will be followed, as closely as possible, when analysing and reporting data.

Any deviations from this SAP will be described and justified in the final report or publication of the trial (using a table as shown in Appendix A). The analysis will be carried out by an appropriately qualified statistician, who should ensure integrity of the data during their data cleaning processes.

2. Background and rationale

The background and rationale for the trial are outlined in detail in the protocol. In brief, surgical site infection (SSI) is a preventable post-operative complication that causes significant pain, suffering and even death. SSIs affect at least 200,000 patients in the UK each year. At an average cost of £3500 per infection, SSIs currently cost the NHS approximately £700 million per year. Patients who develop an SSI are twice as likely to die as those without SSI, and around one third of post-operative deaths are attributable, at least in part, to SSI[1]. With antibiotic resistance becoming an increasingly serious global issue, there is significant health need for high quality research in this area, to try to address this potentially avoidable complication and thereby benefit both patients and the NHS.

Negative pressure wound therapy (NPWT) has been used since the mid-1990s to manage and promote healing in chronic open wounds. The closed negative-pressure environment serves to prevent bacterial ingress and removes blood and serous fluid exuding from the wound. Single-use Negative Pressure Dressings (SUNPD) have been developed for use on primarily closed incisions. These devices are small, highly portable and simple to apply. Their use in studies of non-abdominal wall incisions has resulted in a 50-80% reduction in SSI. However, no multi-centre or randomised trials exploring the clinical effectiveness of SUNPD in reducing SSIs in patients undergoing abdominal surgery have been reported.

3. Trial objectives

The primary objective is to determine if the use of a SUNPD in adult patients undergoing emergency laparotomy reduces SSI at 30 days compared to surgeon's preference of dressing (which may be conventional occlusive dressings, skin glue or no dressing but not another SUNPD).

Secondary objectives are to determine if:

- The use of SUNPD reduces length of hospital stay after surgery
- The use of SUNPD reduces the rate of wound complications
- The use of SUNPD reduces wound complication related hospital re-admission rates
- The use of SUNPD improves health related quality of life
- The use of SUNPD is safe in this population
- The use of SUNPD is acceptable to patients and healthcare professionals
- The use of SUNPD is cost-effective compared to the use of the surgeon's preference of dressings.

4. Trial methods

4.1. Trial design

SUNRRISE is an international, multi-centre, pragmatic, phase lll randomised controlled trial with internal feasibility study comparing the use of SUNPD with the surgeon's preference of dressing (which may be conventional occlusive dressings, skin glue or no dressing but not another SUNPD) after emergency laparotomy.

Patients will be recruited from hospitals undertaking emergency abdominal surgery in the UK and Australia.

See Appendix B for trial schema.

4.2. Trial interventions

The trial intervention is the application of a SUNPD. The SUNPD will be left in-situ until discharge from hospital or until day 7 post-operatively (whichever is sooner).

The control is the surgeon's preference of dressing (which may be conventional occlusive dressings, skin glue or no dressing but not another SUNPD). These will be left in-situ according to local practice.

4.3. Primary outcome measure

The primary outcome is SSI within 30 days of surgery, as defined by the internationally accredited Centers for Disease Control and Prevention (CDC) criteria.

<u>The following CDC definition will be used to identify an SSI</u>:

- The infection must occur within 30-days of the index operation

<u>AND the patient must have at least one of the following</u>:

- Purulent drainage from the wound
- Organisms are detected from a wound swab
- Wound opened spontaneously or by a clinician AND, at the surgical wound, the patient has at least one of:
  - Pain or tenderness
  - Localised swelling
  - Redness
  - Heat
  - Systemic fever (>38ºC)
- Diagnosis of SSI by a clinician or on imaging

4.4. Secondary outcome measures

<u>The secondary outcomes are as follows</u>:

- Length of hospital stay after surgery measured from the date of index surgery to the date of discharge (this will be reported for UK patients only).
- Wound complications within 30 days post-surgery as graded by Clavien-Dindo scale.
- Hospital re-admission for wound related complications within 30 days. These will include SSI's, wound breakdown/dehiscence, seromas and wound related pain.
- Health-related Quality of Life assessed using the validated Short Form-12 (SF-12) questionnaire at baseline, and day 7 and day 30, and the EQ-5D-5L (5 level) at baseline and day 7, day 14, day 21 and day 30.
- Patient reported pain at the site of primary laparotomy assessed using a 10-point visual analogue scale of 1 (no pain at all) - 10 (worst possible pain) at day 7 and day 30.
- Serious adverse events up to 30 days.
- Cost-effectiveness assessed using a patient diary for patient reported healthcare resource usage. Healthcare usage will be taken directly from the patient diaries and the costs attributable to this will be identified.
- Patient acceptability of the use of their dressing via an acceptability score using a 10-point visual analogue scale of 1 (completely acceptable) - 10 (totally unacceptable) at day 7, reflecting patient's assessment of the acceptability of having the dressing.
- In the UK only, healthcare professional's acceptability of use of SUNPD (via a survey of users). This will focus on the ease of application of the dressing, the care needed to maintain/monitor the dressing while it is in place and an overall assessment of health professional's experience of the dressing.

4.5. Timing of outcome assessments

The schedule of trial procedures and outcome assessments are given in Appendix C.

4.6. Randomisation

Randomisation will be provided by a secure online randomisation system and an automated telephone randomisation system managed by a third party (CHaRT, Health Services Research Unit at the University of Aberdeen).

Patients will be randomised in theatre by the operative team following commencement of closure of skin. Patients will be randomised at the level of the individual in a 1:1 ratio to either SUNPD or surgeon's preference of dressing.

A minimisation algorithm will be used within the randomisation system to ensure balance in the treatment allocation over the following variables:

- **Degree of contamination**
    - Clean
    - Clean contaminated
    - Contaminated
    - Dirty

- **Whether or not a stoma is present**
    - Yes (pre-existing/formed*)
    - No

** If a stoma is present, whether it was pre-existing or formed during the operation will also be ascertained. However, this distinction will not be factored in to the minimisation process.*

- **Recruiting site**

So that the randomisation process is not completely deterministic, a 'random element' will be included in the minimisation algorithm, so that each patient has a probability (unspecified here) of being randomised to the opposite treatment that they would have otherwise received.

80

4.7. Sample size

The justification for the sample size is based on data from the ROSSINI trial[2], which reported an SSI rate of 25% in the control group.

To detect a relative reduction of 40% in SSI rates (i.e. from 25% to 15%, so 10% absolute difference) between groups using the standard method of difference between proportions (2-sided) with 90% power and a type I error rate of 5% (i.e. $\alpha=0.05$), requires 336 participants per group to be randomised, so 672 in total. Assuming and adjusting for a 20% attrition/loss to follow-up rate (based on the death rate in this population being approximately 10% at 30 days; further drop out 10%), 840 participants (420 per group) will need to be recruited. Stata 15 software was used to compute the sample size calculation using the two-sample proportions test.

The 40% reduction correlates with the relative reduction being assessed in other large HTA funded SUNPD trials, such as WHIST.

4.8. Framework

The objective of the trial is to test the superiority of SUNPD versus control (surgeon's preference of dressing).

The null hypothesis is that there is no difference in SSI within 30 days of surgery between the intervention groups. The alternative hypothesis is that there is a difference between the groups.

4.9. Interim analyses and stopping guidance

A separate Data Monitoring Committee (DMC) reporting template will be drafted and agreed by the DMC including an agreement on which outcomes will be reported at interim analyses. The statistical methods stated in this SAP will be followed for the outcomes included in the DMC report, where possible.

Data analyses will be supplied in confidence to the independent DMC, which will be asked to give advice on whether the accumulated data from the trial, together with the results from other relevant research, justifies the continuing recruitment of further patients. The DMC will operate in accordance with the SUNRRISE Trial specific charter based upon the template created by the Damocles Group.

The DMC will meet on an annual basis unless there is a specific reason to amend the schedule. It convened at the end of the feasibility phase to review the safety data. Additional meetings may be called if recruitment is much faster than anticipated and the DMC may, at their discretion, request to meet more frequently or continue to meet following completion of recruitment. An emergency meeting may also be convened if a safety issue is identified. The DMC will report directly to the Trial Steering Committee (TSC) who will convey the findings of the DMC to the TMG. The DMC will oversee the data from both the UK and Australian sites.

If participants randomised to the SUNPD arm are doing overwhelmingly better or worse than those not randomised to this group (i.e. control arm) with respect to SSI rates at 30 days, then this effect may become apparent before the target recruitment has been reached. Alternatively, new evidence could emerge from other sources to suggest that SUNPD is definitely more, or less, effective than control. To protect against any unnecessary continuance of the trial in this event, interim analyses of major endpoints and safety data will be supplied during the period of recruitment to the study, in strict confidence, to the DMC along with updates on results of other related studies, and any other analyses that the DMC may request.

The DMC will advise the chair of the TSC if, in their view, any of the randomised comparisons in the trial have provided both:

a) proof beyond reasonable doubt that for all, or for some, types of patient one particular intervention is definitely indicated or definitely contra-indicated in terms of a net difference of a major endpoint, and
b) evidence that might reasonably be expected to influence the patient management of many clinicians who are already aware of the other main trial results.

Unless this happens, however, the TSC, the collaborators and all of the central Trial staff (except the statisticians who supply the confidential analyses) will remain ignorant of the interim results.

Appropriate criteria of proof beyond reasonable doubt cannot be specified precisely, but a difference of at least $p<0.001$ (similar to a Haybittle-Peto stopping boundary) in an interim analysis of a major endpoint may be required to justify halting, or modifying, the study prematurely. If this criterion were to be adopted, it would have the practical advantage that the exact number of interim analyses would be of little importance, so no fixed schedule is proposed.

Given the proposed use of the Haybittle-Peto boundary, no adjustment for multiple testing (to control the overall type I error rate) is proposed, i.e. the threshold for statistical significance at final analysis will still be $p=0.05$.

81

4.10. Internal Feasibility Study Progression Rules

The success of the feasibility study was based upon three factors:

- Patient recruitment
- Adherence to the intervention
- Study drop out

The number of patients to be recruited in the six-month internal feasibility period was 70 patients from at least 5 recruiting sites. The decision for the trial to continue was decided by pre-defined stop-go criteria. A traffic light system was used to determine continuation to the full trial.

- **Green**: recruitment rate >70%, adherence rate >80%, and drop-out rate <20%.
  - If all three criteria are met; continue the trial with protocol unchanged.
- **Amber**: recruitment rate 50-70%, adherence rate 50-80%, or drop-out rate 20-35%.
  - If one or more of our amber criteria are met, then the study will need review to see what changes (if any) could be made to improve whichever criteria are not at the "green" level.
- **Red**: recruitment rate <50%, adherence rate <50%, or drop-out rate >35%
  - If one or more of these criteria are met, we will discuss with the TSC and the funder regarding feasibility of the study continuing.

<u>The definition of the factors used in the stop-go criteria are</u>:

- Recruitment rates: proportion of recruitment target achieved (aim was to recruit 70 patients).
- Successful adherence to the trial arm allocated: successful application of the appropriate dressing for at least 24 hours.
- Study drop-out: complete withdrawal from the study, with no further data to be collected from the patient.

In addition to these stop-go criteria, there was an assessment of safety by the DMC at the end of the feasibility phase. The decision regarding continuing the randomised controlled study was made by the TSC based on both: the traffic light criteria (above); and confirmation from the DMC that there were no safety issues to prevent the trial from continuing.

4.11. Timing of final analysis

The final analysis for the study will occur once all participants have completed the Day 30 assessment and the corresponding outcome data has been entered onto the study database and validated as being ready for analysis.

4.12. Timing of other analyses

Not applicable.

4.13. Trial comparisons

All references in this document to 'group' refer to SUNPD or Control.

5. Statistical Principles

5.1. Confidence intervals and p-values

All estimates of differences between groups will be presented with two-sided 95% confidence intervals, unless otherwise stated. P-values will be reported from two-sided tests at the 5% significance level.

5.2. Adjustments for multiplicity

No correction for multiple testing will be made.

5.3. Analysis populations

All primary analyses (primary and secondary outcomes including safety outcomes) will be by intention-to-treat (ITT). Participants will be analysed in the intervention group to which they were randomised, and all participants shall be included whether or not they received the allocated intervention. This is to avoid any potential bias in the analysis. As a sensitivity analysis, a per-protocol analysis will also be carried out for the primary outcome. See section 5.4 for definition of adherence and section 9.10 for further details on sensitivity analyses.

5.4. Definition of adherence

For participants randomised to the SUNPD arm, they will be deemed adherent to their randomised allocation if they received the SUNPD and the dressing was on for at least 3 days or until day of discharge if discharged earlier than 3 days. For the control arm, participants will be deemed adherent if they do not receive the SUNPD and receive the surgeon's preference of dressing (which may be conventional occlusive dressings, skin glue or no dressing but not another SUNPD). This definition of adherence will be used to define the per-protocol set.

We may also consider a more stringent definition for adherence to SUNPD as a sensitivity analysis where participants will be deemed adherent if they received the SUNPD and the dressing was on for at least 5 days or until day of discharge if discharged earlier than 5 days.

5.5. Handing protocol deviations

A protocol deviation is defined as a failure to adhere to the protocol such as errors in applying the inclusion/exclusion criteria, the incorrect intervention being given, incorrect data being collected or measured, follow-up assessments outside the assessment window or missed follow-up assessments. We will apply a strict definition of the ITT principle and will include all participants as per the ITT population described in section 5.3 in the analysis, in some form, regardless of deviation from the protocol.[3]

Time window:
The primary outcome and most of the secondary outcomes are defined to be within the 30 days post-surgery time-period. However, there may be instances where for some participants, their day 30 follow up assessment form is completed after day 30 post-surgery, i.e. completed late, and so a +14 day time window for the 30 day follow up assessment is specified in the protocol. To protect against this, the 30 day form instructs that the responses provided are to relate to the 30 day post-surgery time period, so that even if the assessment was conducted/completed after 30 days post-surgery, all data are related to the 30 day period following the surgery for the patient.

There may also be instances where for some participants, their day 30 follow up assessment is done before day 30 post-surgery, i.e. done earlier than expected but there is no time-window in the protocol for any assessments being done early. To comply with the ITT principle, for the ITT analysis for the primary outcome and for all secondary outcomes, the time window will not be considered and so data from all 30 day follow up forms completed will be included in the analysis.

As sensitivity analyses, the analysis for the primary outcome and for any secondary outcomes (where appropriate) will be conducted as following (see section 9.10 for further details):

- Excluding any 30 day forms completed before day 30 post-surgery but include any 30 day forms completed after day 30 post-surgery
- Only including 30 day forms completed within 30-44 days (i.e. +14 day time-window) post-surgery i.e. exclude any 30 day forms completed before day 30 post-surgery or after day 44 post-surgery.

5.6. Unblinding

It is not feasible to blind either the participants or the inpatient wound assessors (day 7) to treatment allocation because the presence of the negative pressure dressing is clearly identifiable. The Day 30 wound review will be undertaken by a trained wound assessor who will be blind to treatment allocation and who has not previously been involved in the care of the patient. The importance of blinding will be explained to participants and they will be asked to not inform the wound assessor of their treatment arm.

6. Trial population

6.1. Recruitment

A flow diagram (as recommended by CONSORT[4]) will be produced to describe the patient flow through each stage of the trial. This will include information on the number (with reasons) of losses to follow-up (drop-outs and withdrawals) over the course of the trial. A template for reporting this is given in Appendix D1.

6.2. Baseline characteristics

The trial population and operative details will be tabulated as per Appendix D2 and D3. Categorical data will be summarised by number of participants, counts and percentages. Continuous data will be summarised by the number of patients, mean and standard deviation if deemed to be normally distributed or number of participants, median and interquartile range if data appear skewed, and ranges if appropriate. Tests of statistical significance will not be undertaken, nor confidence intervals presented.[5]

7. Intervention(s)

7.1. Description of the intervention(s)

The trial intervention is the application of a SUNPD. The Smith and Nephew SUNPD is a CE-marked device, Therapeutic Goods Administration approved, and available for routine clinical use throughout the UK and Australia. It will be used within its licensed indication in the SUNRRISE Trial. The SUNPD will be left in situ until discharge from hospital or until day 7 post-operatively (whichever is sooner). The dressing should not be routinely changed but if, for a clinical reason, the dressing needs to be changed, it should be replaced with another SUNPD.

The control is the surgeon's preference of dressing (which may be conventional occlusive dressings, skin glue or no dressing but not another SUNPD). These will be left in situ according to local practice. When conventional occlusive dressings are used, it is expected that most participants will have this changed in hospital and discharged with a dressing in place.

7.2. Adherence to allocated intervention

A cross-tabulation of allocated intervention by the adherence categories stated in section 5.4 will be produced (proportions and percentages). A template for reporting adherence is given in Appendix D4.

82

8. Protocol deviations

Frequencies and percentages by group will be tabulated for the protocol deviations as per Appendix D5.

9. Analysis methods

Intervention groups will be compared using appropriate statistical models to adjust for all covariates as specified in section 9.1, where possible. See section 9.5 - 9.10 which describes in detail for each outcome the type of analysis method to be used.

9.1. Covariate adjustment

In the first instance, intervention effects between groups for all outcomes will be adjusted for the minimisation parameters listed in section 4.6. Centre will be included as a random effect term in the model, and all other factors as fixed effects. Other covariate adjustment will be baseline values for parameters where available (e.g. the analysis of SF-12 and EQ-5D-5L) will include the baseline value as a covariate in the model. The control arm will be the reference category for all analyses. If the log-binomial model fails to converge, a Poisson regression model with robust standard errors will be used to estimate the same parameters[6]. If this also fails to converge, then the log-binomial model without the random effect term for centre will be fitted. If we still observe issues with convergence in the model, then unadjusted estimates will produced from the log-binomial model. It will be made clear in the final report why this occurred (e.g. not possible due to low event rate/lack of model convergence).

9.2. Distributional assumptions and outlying responses

Distributional assumptions (e.g. normality of regression residuals for continuous outcomes will be assessed visually prior to analysis; although in the first instance the proposed primary method of estimation in this analysis plan will be followed. If responses are considered to be particularly skewed and/or distributional assumptions violated, the impact of this will be examined through sensitivity analysis; this will consist of 1) transformation of responses prior to analysis (e.g. log transformation for continuous outcomes) or 2) use of an appropriate non-parametric method of analysis.

9.3. Handling missing data

In the first instance, analysis will be completed on received data only with every effort made to follow-up participants to minimise any potential for bias. To examine the possible impact of missing data on the results, and to make sure we are complying with the ITT, sensitivity analysis will be performed on the primary outcome measure.[7] See section 9.10 for further details.

9.4. Data manipulations

Age (years)
(Randomisation date – Date of birth) / 365.25.

BMI
If BMI is missing then BMI will be computed from the participants height and weight using the following formula: BMI = Weight (kg) / (Height (m))^2

This will then be categorised into 4 categories:
Underweight (BMI<18.5), Healthy weight (BMI 18.5-24.9), Overweight (BMI 25.0-29.9), Obese (BMI ≥30.0)

Operative procedure performed
There are 30 options for this:
1) Adhesiolysis, 2) Appendicectomy, 3) Cholecystectomy,
4) Colectomy: left (including anterior resection), 5) Colectomy: right, 6) Colectomy: subtotal,
7) Colorectal resection - other, 8) Colostomy formation/revision,
9) Drainage of abscess/collection, 10) Exploratory laparotomy only,
11) Gastric surgery – other, 13) Hartmann's procedure, 14) Ileostomy formation/ revision,
15) Intestinal bypass, 16) Peptic ulcer – over sew of bleed,
17) Peptic ulcer – suture or repair of perforation, 18) Repair of intestinal perforation,
19) Resection of other intraabdominal malignancy, 20) Small bowel resection, 21) Transplant,
22) Trauma, 23) Vascular procedure, 24) Washout only, 12) Other, 25) Hernia,
26) Intestinal bypass – involving colon, 27) Intestinal bypass – not involving colon,
28) Drainage of abscess/collection – related to colon,
29) Drainage of abscess/collection – related to small bowel/stomach,
30) Drainage of abscess/collection – not related to bowel

Each procedure has been assigned to one of these groups:

Procedure	Sub-group
2) Appendicectomy	Bowel – colonic
4) Colectomy: left (including anterior resection)	Bowel – colonic
5) Colectomy: right	Bowel – colonic
6) Colectomy: subtotal	Bowel – colonic
7) Colorectal resection - other	Bowel – colonic
8) Colostomy formation/revision	Bowel – colonic
13) Hartmann's procedure	Bowel – colonic
26) Intestinal bypass – involving colon	Bowel – colonic
28) Drainage of abscess/collection – related to colon	Bowel – colonic
11) Gastric surgery – other	Bowel - non-colonic
14) Ileostomy formation/ revision	Bowel - non-colonic

16) Peptic ulcer – over sew of bleed	Bowel - non-colonic
17) Peptic ulcer – suture or repair of perforation	Bowel - non-colonic
18) Repair of intestinal perforation	Bowel - non-colonic
20) Small bowel resection	Bowel - non-colonic
27) Intestinal bypass – not involving colon	Bowel - non-colonic
29) Drainage of abscess/collection – related to small bowel/stomach	Bowel - non-colonic
1) Adhesiolysis	Non-bowel
3) Cholecystectomy	Non-bowel
10) Exploratory laparotomy only	Non-bowel
21) Transplant	Non-bowel
22) Trauma	Non-bowel
23) Vascular procedure	Non-bowel
24) Washout only	Non-bowel
12) Other	Non-bowel
25) Hernia	Non-bowel
30) Drainage of abscess/collection – not related to bowel	Non-bowel
9) Drainage of abscess/collection	n/a
15) Intestinal bypass	n/a
19) Resection of other intraabdominal malignancy	n/a

The options will be collapsed into 3 categories:
Bowel – colonic, Bowel – non-colonic, and Non-bowel

Where there are multiple procedures listed within a single operation, they will be coded as following:

- Bowel – colonic = at least one procedure of that type
- Bowel – non-colonic = at least one procedure of that type and no "Bowel – colonic" procedures
- Non-bowel = at least one procedure of that type and no "Bowel – colonic" or "Bowel – non-colonic" procedures

Any operation that does not have procedures that allow categorisation will not be included.

Size of Wound
This is a continuous variable which will be categorised into 2 categories:
1) <15cm, 2) ≥15cm

Skin Preparation used
There are 7 categories for this:
1) Aqueous betadine, 2) Alcoholic betadine, 3) 0.5% Aqueous Chlorhexidine, 4) Other
5) 0.5% Alcoholic Chlorhexidine, 6) 2% Aqueous Chlorhexidine, 7) 2% Alcoholic Chlorhexidine

This variable will be collapsed into 3 categories:
1) 2% Alcoholic Chlorhexidine, 2) Aqueous betadine, 3) All other skin prep

Primary outcome SSI according to the CDC criteria

- The infection must occur within 30-days of the index operation

AND the patient must have at least one of the following:

- Purulent drainage from the wound
- Organisms are detected from a wound swab
- Wound opened spontaneously or by a clinician AND, at the surgical wound, the patient has at least one of:
  - *a) Pain or tenderness*
  - *b) Localised swelling*
  - *c) Redness*
  - *d) Heat*
  - *e) Systemic fever (>38 degrees C)*
- Diagnosis of SSI by a clinician or on imaging

This data is collected in the "Wound Assessment Day 7 or on Discharge (if sooner)" case report form (CRF), "Wound Assessment on Day 30" CRF and "Return To Theatre Form". To cover the interim period between "Day 7 or Discharge" and "Day 30", the participants are asked to complete daily diaries. In the Patient Diary, there is one question that is related to the primary outcome "1) I have been prescribed/I am taking antibiotics for my wound".

<u>The primary outcome will be coded as a binary (Yes/No), where it will be</u>:

- Yes, if patient has had a SSI as per the CDC criteria from the Day-7/Discharge, Day-30, or Return to Theatre forms, or if the patient has been prescribed/is taking antibiotics for their wound as reported from the patient diary within 30 days of the index operation.
- No, if the patient has completed the 30-day assessment and has not had a SSI as per the CDC criteria at any point and they have not been prescribed/have not taken antibiotics for their wound as reported from the patient diary within 30 days of the index operation.

<u>Note</u>: Any patient diaries that are completed before date of discharge or on the day of discharge will not be used for any analysis since the diaries were meant to cover the interim period from discharge and final assessment at day 30. In addition, if the patient returns to theatre after their day 30 assessment, then this form will not be used for any analysis.

Length of hospital stay (days)

Date of discharge – date of surgery. This is calculated and reported for UK patients only.

<u>Note</u>: if date of discharge is the same as date of surgery, then length of stay will be 0 days.

Wound complications within 30 days post-surgery

The data for other wound complications is collected in the "Wound Assessment Day 7 or on Discharge (if sooner)" CRF and "Wound Assessment on Day 30" CRF. There is a question on both forms that collects this data; 1) Has there been any other wound complications (excluding wound infection).

This outcome will be coded as a binary Yes/No, where it will be:

- Yes, if the patient reports any instances of wound complications (excluding wound infection) as recorded on any of the forms mentioned above.
- No, if the patient has completed the 30-day assessment and does not report any instances of wound complications (excluding wound infection) on any of the forms mentioned above.

Clavien-Dindo classification

The data for management of wound infections and other wound complications is collected in the "Wound Assessment Day 7 or on Discharge (if sooner)" CRF and "Wound Assessment on Day 30" CRF. There are questions on both forms that collects this data; 1) If the patient had a wound infection, what management did they receive, and 2) If the patient had any other wound complication (excluding wound infection), what management did they receive.

The Clavien-Dindo classification system grades complications according to the management, treatment and/or therapy used to address it.

This data will be coded as below:

Grade	Management of wound infections and other wound complications
I	• None / conservative management • On ward intervention
II	• Antibiotic drug treatment
III	• Surgical intervention • Radiological intervention
IV	• ITU admission
V	n/a – death and associated details reported using SUNRRISE SAE form

Hospital readmission for wound related complications within 30 days

The data for this outcome is collected on the "Wound Assessment on Day 30" CRF, and there are two questions on this form that relate to this outcome; 1) If the patient had a wound infection, were they re-admitted to hospital as a result?, and 2) If the patient had any other wound complications, were they re-admitted to hospital as a result?

This outcome will be coded as a binary Yes/No, where it will be:

- Yes, if the patient is re-admitted because of any wound related complications as mentioned above.
- No, if the patient has completed the 30-day assessment and has not been re-admitted because of wound related complications.

SF-12

The SF-12 questions will be coded as follows:

SFQ1 – GENERAL HEALTH
Excellent=5
Very good=4.4
Good=3.4
Fair=2
Poor=1

SFQ2a – MODERATE ACTIVITIES
Yes, limited a lot=1
Yes, limited a little=2
No, not limited at all=3

SFQ2b – CLIMBING STAIRS
Yes, limited a lot=1
Yes, limited a little=2
No, not limited at all=3

SFQ3a – ACCOMPLISHED LESS DUE TO PHYSICAL HEALTH
All of the time=1
Most of the time=2
Some of the time=3
A little of the time=4
None of the time=5

SFQ3b – LIMITED IN WORK OR OTHER ACTIVITIES DUE TO PHYSICAL HEALTH
All of the time=1
Most of the time=2
Some of the time=3
A little of the time=4
None of the time=5

SFQ4a – ACCOMPLISHED LESS DUE TO EMOTIONAL PROBLEMS
All of the time=1
Most of the time=2
Some of the time=3
A little of the time=4
None of the time=5

SFQ4b – DID WORK/ACTIVITIES LESS CAREFULLY DUE TO EMOTIONAL PROBLEMS
All of the time=1
Most of the time=2
Some of the time=3
A little of the time=4
None of the time=5

SFQ5 – PAIN INTERFERED WITH NORMAL WORK
Not at all=5
A little bit=4
Moderately=3
Quite a bit=2
Extremely=1

SFQ6a – HAVE YOU FELT CALM AND PEACEFUL
All of the time=5
Most of the time=4
Some of the time=3
A little of the time=2
None of the time=1

SFQ6b – DID YOU HAVE A LOT OF ENERGY
All of the time=5
Most of the time=4
Some of the time=3
A little of the time=2
None of the time=1

SFQ6c – HAVE YOU FELT DOWNHEARTED AND LOW
All of the time=1
Most of the time=2
Some of the time=3
A little of the time=4
None of the time=5

SFQ7 – INTERFERED WITH SOCIAL ACTIVITES
All of the time=1
Most of the time=2
Some of the time=3
A little of the time=4
None of the time=5

The following domains will be computed from the SF-12 questionnaire:

- Physical Function (PF) = SFQ2a + SFQ2b
  - Physical Function Score = 100*((PF-2)/4)
- Role Limitation Due to Physical Problems (RP) = SFQ3a + SFQ3b
  - Role Limitation Due to Physical Problems score = 100*((RP-2)/8)
- Role Limitation Due to Emotional Problems (RE) = SFQ4a + SFQ4b
  - Role Limitation Due to Emotional Problems Score = 100*((RE-2)/8)
- Social Functioning (SF) = SFQ7
  - Social Functioning Score = 100*((SF-1)/4)
- Mental Health (MH) = SFQ6a + SFQ6c
  - Mental Health Score = 100*((MH-2)/8)
- Vitality (VT) = SFQ6b
  - Vitality Score = 100*((VT-1)/4)
- Pain (P) = SFQ5
  - Pain Score = 100*((P-1)/4)
- General Health Perception (GHP) = SFQ1
  - General Health Perception Score = 100*((GHP-1)/4)

Formulas for z-score standardization of SF-12 scales:

PF_Z = (PF – 81.18122) / 29.10558

RP_Z = (RP – 80.52856) / 27.13526

BP_Z = (BP – 81.74015) / 24.53019

GH_Z = (GH – 72.19795) / 23.19041

VT_Z = (VT – 55.59090) / 24.84380

SF_Z = (SF – 83.73973) / 24.75775

RE_Z = (RE - 86.41051) / 22.35543

MH_Z = (MH – 70.18217) / 20.50597

Formulas for aggregating scales in estimating aggregate physical and mental summary scores:

- **AGG_PHYS**

(PF_Z * 0.42402) + (RP_Z * 0.35119) + (BP_Z * 0.31754) + (GH_Z * 0.24954) + (VT_Z * 0.02877) + (SF_Z * -0.00753) + (RE_Z * -0.19206) + (MH_Z * -0.22069)

- **AGG_MENT**

(PF_Z * -0.22999) + (RP_Z * -0.12329) + (BP_Z * -0.09731) + (GH_Z * -0.01571) + (VT_Z * 0.23534) + (SF_Z * 0.26876) + (RE_Z * 0.43407) + (MH_Z * 0.48581)

Formulas for t-score transformation of summary scores for PCS and MCS scores:
Transformed Physical (PCS) = 50 + (AGG_PHYS * 10)
Transformed Mental (MCS) = 50 + (AGG_MENT * 10)

Note: If any question is missing a response then computation of the PCS and MCS cannot be calculated.

EQ-5D (5 level)

The current NICE guidelines (updated October 2019) on the use of EQ-5D-5L scoring based on the most recent value set for England published by Devlin et al. 2018 advise not to use this and instead to map the 5L data onto the 3L value set based on a mapping function developed by van Hout et al. 2012.

EQ-5D-5L have developed the crosswalk value sets for the 5L to 3L, and so these values will be used for scoring (https://euroqol.org/eq-5d-instruments/eq-5d-5l-about/valuation-standard-value-sets/crosswalk-index-value-calculator/).

Note: If any of the 5 questions is missing a response then computation for EuroQol score cannot be calculated.

Also for those patients that die, for the Index score a value of "0" will be imputed for all subsequent time-points post their death since for this questionnaire, a value of 0=death.

Serious adverse events (SAEs) up to 30 days

This data is collected in the "Wound Assessment Day 7 or on Discharge (if sooner)" CRF, "Wound Assessment on Day 30" CRF and the "Serious Adverse Event (SAE) form", where each SAE will be counted as a separate event from the SAE form.

The questions related to SAEs on the "Day 7 or discharge" form are:

1) "If the patient had a wound infection, did it prolong their hospitalisation?"
2) "If the patient had any other wound complications, did it prolong their hospitalisation?"
3) "Has the patient had any of the following complications following surgery?"
   a) An anastomotic leak
   b) An intra-peritoneal collection
   c) A thrombo-embolic event
   d) An infection not related to wound
   e) A cardiac or central nervous system complication
   f) Paralytic ileus

The questions related to SAEs on the "Day 30" form are:

1) "If the patient had a wound infection, was their initial hospitalisation prolonged?"
2) "If the patient had a wound infection, were they re-admitted to hospital?"
3) "If the patient had any other wound complications, was their initial hospitalisation prolonged?"
4) "If the patient had any other wound complications, were they re-admitted to hospital?"
5) "Has the patient had any of the following complications following surgery?"
   a) An anastomotic leak
   b) An intra-peritoneal collection
   c) A thrombo-embolic event
   d) An infection not related to wound
   e) A cardiac or central nervous system complication
   f) Paralytic ileus

This outcome will be computed as a binary (Yes/No) where "yes" will be participants that experienced any SAE within 30 days post-surgery and "no" will be participants that experienced no SAE within 30 days post-surgery. Since participants can have more than one SAE, a tabulation of the number of SAE's each participant has experienced will be provided. Patients that have experienced no SAE's will be reported as "0".

Note: Since the question "Has the patient had any of the following complications following surgery?" from the day 7 and day 30 forms are the same, it is likely that if the patient has experienced an event (e.g. anastomotic leak) at day 7 and at day 30, then it is likely to be the same event, but just recorded twice. Similarly the questions related to prolonging hospitalisation due to wound infection or other complications are likely to be the same between day 7 and day 30 form. Therefore to avoid double counting, when an event from these specific questions are recorded on both the day 7 and day 30 forms, this will be counted as one event.

83

9.5. Analysis methods – primary outcome(s)

The primary outcome is SSI within 30 days of surgery, as defined by the internationally accredited CDC criteria.

This outcome is a binary outcome (i.e. yes/no) and will be computed based on details provided in the data manipulation section 9.4. The number and percentage of participants reporting an SSI within 30 days of index surgery will be reported by treatment group. An adjusted relative risk and 95% Confidence Interval (CI), as well as the adjusted risk difference and 95% CI will be estimated from a mixed effects binomial regression model. A log link will be used in the model to estimate the relative risk and the identity link will be used in the model to estimate the risk difference. The p-value for the adjusted relative risk will be reported.

A template for reporting the primary outcome is given in Appendix D6.

9.6. Analysis methods – secondary outcomes

A template for reporting all the secondary outcomes is given in Appendix D7.

- **Length of hospital stay after surgery**

This outcome will be treated as continuous data, and will be summarised using means and standard deviations. It will be analysed using a mixed effects linear regression model to estimate the adjusted mean difference between groups and corresponding 95% CI. Length of hospital stay after index surgery will be computed as the difference (in days) between the date of surgery and date of discharge. For participants that are discharged on the same day as the surgery day, number of days in hospital for these participants will be "0" days.

Note: This outcome will be analysed for patients randomised from the UK only, and so all patients randomised from Australia will be excluded from the analysis for this outcome.

- **Wound complications within 30 days post-surgery as graded by Clavien-Dindo**

This outcome is a binary outcome (yes/no) and will be computed based on details provided in the data manipulation section 9.4. This outcome will be analysed using the same analysis methods as described for the primary outcome in section 9.5.

- **Hospital re-admission for wound related complications within 30 days**

This outcome is a binary outcome (yes/no) and will be computed based on details provided in the data manipulation section 9.4. This outcome will be analysed using the same analysis methods as described for the primary outcome in section 9.5.

- **Short Form-12 (SF-12)**

The data for this outcome is collected at baseline, day 7 and day 30. The data for this outcome is continuous and there are two main components to this questionnaire; the transformed Physical Component Score (PCS) and Mental Component Score (MCS). The computation for both of these scores is described in the data manipulations section 9.4.

Difference between group means and associated 95% CI at the primary time points (day 7 and day 30) will be estimated through the use of a repeated measures[8], mixed-effects linear regression model. All assessment times (baseline and the post-treatment time points) will be included. Parameters allowing for participant, treatment group, baseline score, time and the randomisation minimisation variables will be included (all as fixed effects except centre which will be included as a random effect). Time will be assumed to be a categorical (fixed) variable. To allow for a varying treatment effect over time, a time by treatment interaction parameter will be included in the model. Estimates of the mean differences between groups at the relevant time will be taken from the model including this interaction parameter. A general 'unstructured' covariance structure will be assumed. Robust standard errors 'sandwich estimator'[9] will be incorporated to allow for the possibility of the violation of normality of regression residuals.

<u>Note</u>: The analysis for the quality of life outcome will be done separately for PCS and MCS.

- **EuroQuol-5 Dimension-5 Level (EQ-5D-5L)**

The data for this outcome is collected at baseline, day 7, day 14, day 21 and day 30. The data for this outcome is continuous and there are two main components to this questionnaire; the EQ-5D index score and EQ-5D health score. The computation for the EQ-5D index score is described in the data manipulations section 9.4.

The analysis for this outcome will be analysed using the same analysis methods as described for the secondary outcome SF-12.

<u>Note</u>: The analysis the index score and the health score will be done separately.

- **Pain at the site of primary laparotomy**

The data for this outcome is collected at day 7 and day 30 and is in the form of a 10-point visual analogue scale from 1-10 were 1 indicates no pain at all and 10 indicates worst possible pain. The data for this outcome will be treated as continuous data and will be analysed using the same analysis methods as described for the secondary outcome Length of hospital stay after surgery.

<u>Note</u>: The analysis for day 7 and day 30 will be done separately.

- **Patient and healthcare professional acceptability of the use of their dressing**

The data for this outcome is collected in the form of a 10-point visual analogue scale from 1-10 were 1 indicates completely acceptable and 10 totally unacceptable. The data for this outcome will be treated as continuous data and will be analysed using the same analysis methods as described for the secondary outcome Length of hospital stay after surgery.

Note: The analysis for patient and healthcare professional acceptability score will be done separately.

9.7. Analysis methods – exploratory outcomes and analyses

Any data that does not form a pre-specified outcome will be presented using simple summary statistics by intervention group (i.e. numbers and percentages for binary data and means (or medians) and standard deviations (or inter-quartile ranges) for continuous normal (or non-normal) data.

9.8. Safety data

This outcome (SAEs up to 30 days) will be computed as a binary outcome (i.e. yes/no) and will be computed based on details provided in the data manipulation section 9.4. The number and percentage of participants experiencing at least 1 SAE within 30 days will be reported by treatment group and will be analysed using the same analysis methods as described for the primary outcome in section 9.5.

There may be instances where participants experience more than one SAE, and so this data will also be presented and summarised as count data by treatment group; i.e. number of participants that experienced exactly 1 SAE, 2 SAE's, 3 SAE's, etc.

The number and percentage of participants experiencing any SAEs and Unexpected and Related Serious Adverse Events (URSAE) will be presented by SAE/URSAE type and intervention group. This summary will also include the total number of events for all SAE/URSAE type by treatment arm.

A template for reporting this data is given in Appendix D8.

84

9.9. Planned subgroup analyses

Interpretation of subgroup analysis will be treated with caution (output will be treated as hypothesis generating rather than definitive[10]).

Analysis will be limited to the primary outcome only, and for the following subgroups:

- Degree of contamination
  - Clean, Clean contaminated, Contaminated, Dirty
- Whether or not a stoma is present
  - Yes, No
- Operative procedure
  - Bowel – colonic, Bowel – non-colonic, Non-bowel
- Size of wound
  - <15cm, ≥15cm
- Skin prep
  - 2% Alcoholic Chlorhexidine, Aqueous betadine, All other skin prep
- Country
  - UK, Australia
- BMI
  - Underweight, Healthy weight, Overweight, Obese
- SSI and Day 30 assessment method
  - Wound visualisation in-person, Wound visualisation remotely either by video or photo, Remotely without visualisation
- Date of COVID-19 as per World Health Organization (For UK patients only)
  - Patients randomised before 11/03/2020, Patients randomised after 11/03/2020

The effects of these subgroups will be examined by including the relevant subgroup by treatment group interaction term in the binomial model for each subgroup analysis. Tests of heterogeneity will be presented along with subgroup specific relative risks and corresponding 95% CI's.

A template for reporting is given in Appendix D9.

9.10. Sensitivity analyses

Sensitivity analyses with respect to the primary outcome will consist of:

- Per-protocol analysis including only participants who remained adherent with their randomised treatment allocation (see section 5.4 for definition of adherence).

- Analysis of the ITT population and:
    - Excluding any 30 day forms completed before day 30 post-surgery but include any 30 day forms completed after day 30 post-surgery
    - Only including 30 day forms completed within 30-44 days (i.e. +14 day time-window) post-surgery i.e. exclude any 30 day forms completed before day 30 post-surgery or after day 44 post-surgery
        - *Note: if* participants *have a SSI confirmed from other forms (i.e. Day-7/discharge, return to theatre, patient diary) which are completed within the time-window of the 30 day assessment, then those participants will be included as having had an SSI in this analysis even if the actual Day-30 assessment is done outside the time-window.*

- Analysis of the ITT population excluding any SSI's identified from the patient diary only.

- An analysis to assess the effect of missing responses. An analysis of the ITT analysis population with missing data imputed in the following ways:
    - SSI (worst case) for the SUNPD group and no SSI (best case) for control group
    - No SSI (best case) for the SUNPD group and SSI (worst case) for control group
        - *Note: The primary outcomes is considered missing for patients where the Day-30 assessment is not completed for whatever reason, and the patient has not had a SSI reported on other forms (i.e. Day-7/Discharge, Return to Theatre, Patient Diary).*

- Analysis of the ITT population excluding any patients that return to theatre where the participant has not already had an SSI diagnosed.

10. Analysis of sub-randomisations

Not applicable.

11. Health economic analysis

As indicated in the protocol there will also be an economic analysis. The details of this analysis are documented separately.

12. Statistical software

Statistical analysis will be undertaken in the following statistical software packages: SAS software, version 9.4 (or higher) and/or Stata version 17 (or higher) will be used for all analyses.

13. References

1. Kirkland KB, Briggs JP, Trivette SL, Wilkinson WE, Sexton DJ. The impact of surgical-site infections in the 1990s: attributable mortality, excess length of hospitalization, and extra costs. Infection Control & Hospital Epidemiology 1999;20(11):725-30.

2. Pinkney TD, Calvert M, Bartlett DC, Gheorghe A, Redman V, Dowswell G, et al. Impact of wound edge protection devices on surgical site infection after laparotomy: multicentre randomised controlled trial (ROSSINI Trial). BMJ 2013;347:f4305.

3. Gupta SK. Intention-to-treat concept: A review. Perspect Clin Res. 2011;2(3):109-112.

4. Schulz KF, Altman DG, Moher D, for the CONSORT Group. CONSORT 2010 Statement: updated guidelines for reporting parallel group randomised trials. BMJ. 2010;340:c332.

5. Altman DG, Dore CJ. Randomisation and baseline comparisons in clinical trials. Lancet. 1990;335:149–53.

6. Zou G. A modified Poisson regression approach to prospective studies with binary data. Am J Epidemiol. 2004;159(7):702-6.

7. White IR, Horton NJ, Carpenter J, Pocock SJ. Strategy for intention to treat analysis in randomised trials with missing outcome data. BMJ. 2011;342:d40.

8. Brown H, Prescott R. Applied Mixed Models in Medicine, 2nd Edition. Wiley, 2006.

9. Diggle, P.J., Liang, K.Y. and Zeger, S.L. (1994) Analysis of Longitudinal Data. Oxford University Press, New York.

10. Wand R, Lagakos SW, Ware JH, Hunter DJ, Drazen JM. Reporting of subgroups analyses in clinical trials. NEJM. 2007;357:2189-94.

11. Devlin NJ, Shah KK, Feng Y, Mulhern B, van Hout B. Valuing health-related quality of life: An EQ-5D-5L value set for England. Health Econ. 2018 Jan;27(1):7-22. doi: 10.1002/hec.3564. Epub 2017 Aug 22. PMID: 28833869; PMCID: PMC6680214.

12. Van Hout B, Janssen MF, Feng YS, Kohlmann T, Busschbach J, Golicki D, Lloyd A, Scalone L, Kind P, Pickard AS. Interim scoring for the EQ-5D-5L: mapping the EQ-5D-5L to EQ-5D-3L value sets. Value Health. 2012 Jul-Aug;15(5):708-15. doi: 10.1016/j.jval.2012.02.008. Epub 2012 May 24. PMID: 22867780.

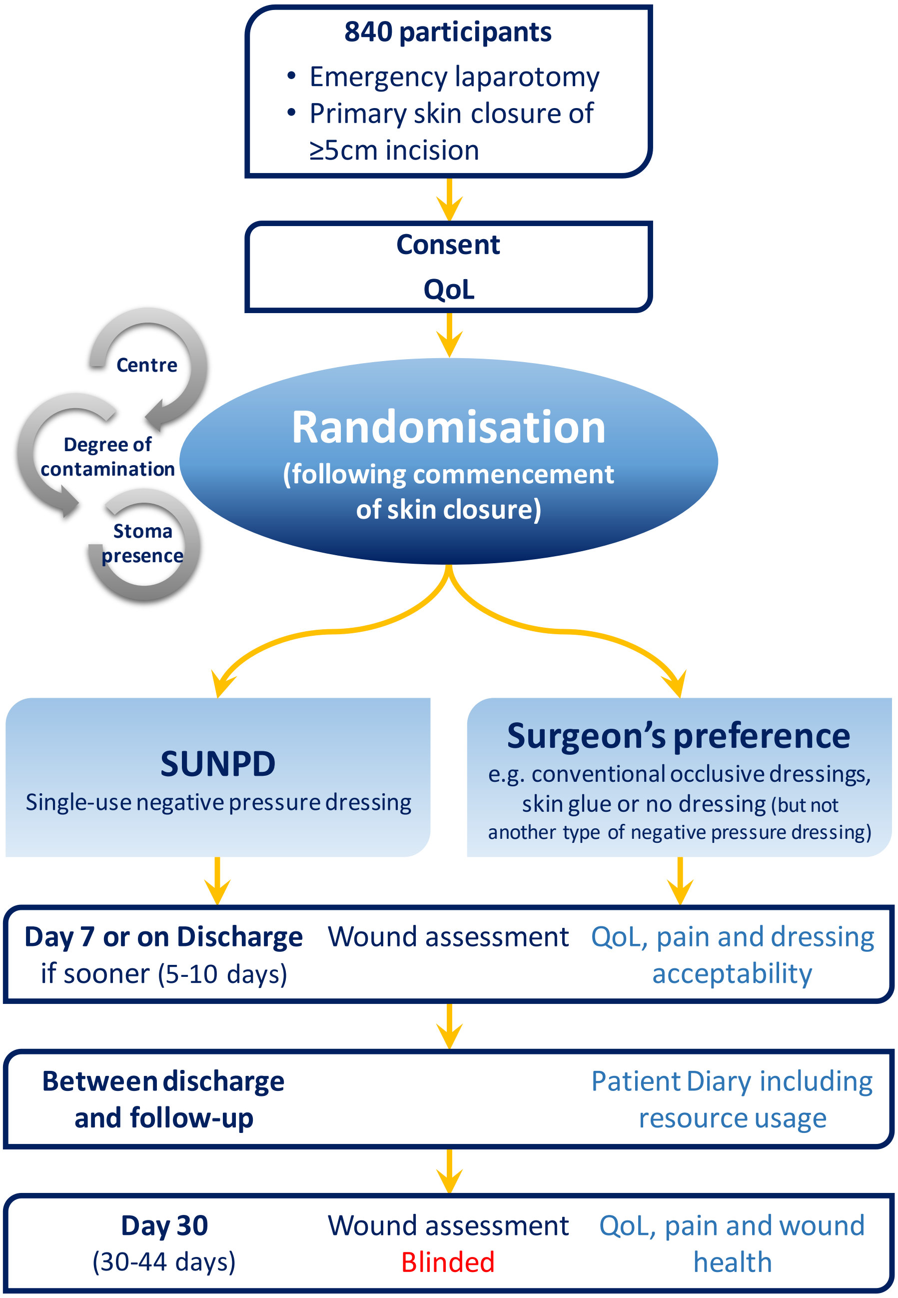
85

Appendix A: Deviations from SAP

This report below follows the statistical analysis plan dated *<insert effective date of latest SAP>* apart from following:

Section of report not following SAP	Reason
<insert section >	<insert, e.g. exploratory analyses request by TMG>

Appendix B: Trial schema



Appendix C: Schedule of assessments

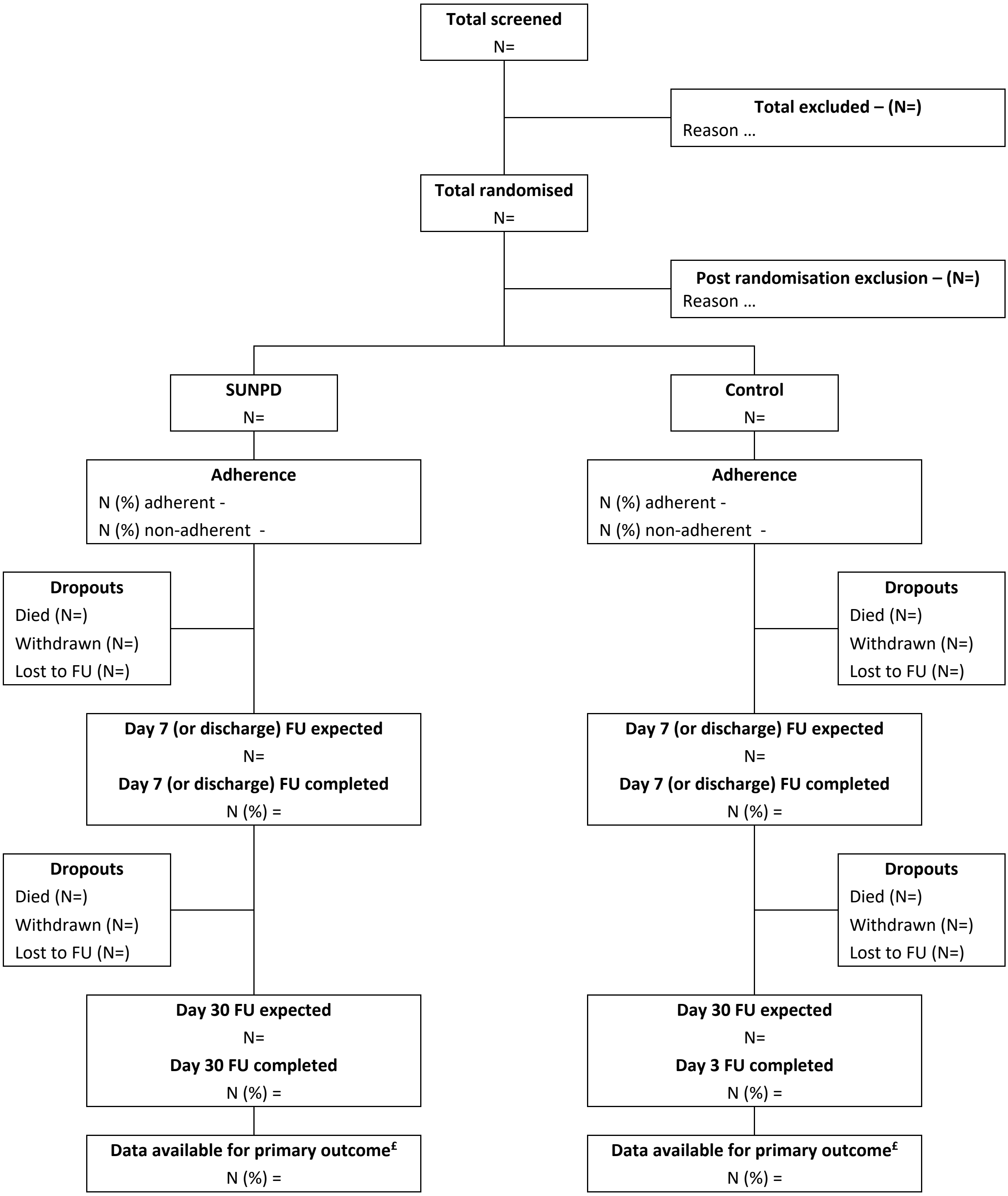
Activity/CRF	Pre-theatre	In-theatre	Day 7 post-op	Day 14 post-op	Day 21 post-op	Day 30 post-op	Day 30+
			(-2/+3 days)	(± 2 days)	(± 2 days)	(+ 14 days)	
Patient identification and screening	On-call surgical team						
Patient consent • Standard	Patient & Member of the research team						
Patient consent • Delayed (UK only)	Consultee/Representative & Member of the research team	Patient & Member of the research team when capacity regained					
Randomisation form	Started pre-theatre by the research team	Surgeon or member of the research team					
In-Theatre form		Ideally an operating surgeon, or member of the research team					
Wound Assessment Day 7 or on Discharge (if sooner)[1,2]			Member of the research team				
EQ-5D-5L	Completed by the participant		Completed by the participant	Completed by the participant	Completed by the participant	Completed by the participant	
SF-12	Completed by the participant		Completed by the participant			Completed by the participant	
Patient diary			Completed daily by the participant following discharge from hospital until they undergo the Day 30 wound review				Patients to continue with a diary if they have an ongoing SSI
Bluebelle wound healing questionnaire						Completed by the participant independently, and then by the participant with a blinded member of the research team reviewing wound	
Wound Assessment Day 30[1]						Completed by a blinded member of the research team as an **in-person or remote (video) review**	
SAE reporting		All serious adverse events by member of the research team using SAE form or wound assessment CRF if excluded from expedited reporting					Related serious adverse events only
Return to theatre form		Member of the research team for any return to theatre following patient returning to theatre					
PI Declaration form for CFR data							Completed by PI at the end of each participant's involvement

[1] Assessment of pain undertaken by a member of the research team by asking the participant and recording the response on the CRF

[2] Score of patient acceptability of dressing undertaken by a member of the research team by asking the participant and recording the response on the CRF

87

Appendix D1: CONSORT flow diagram



£-ITT analysis population for the primary outcome

88

Appendix D2: Baseline characteristics

Baseline Data	SUNPD (N=)	Control (N=)	Total (N=)
Demographic data			
Age at randomisation (years)			
Mean [SD]			
Min ; Max			
Gender			
Female			
Male			
BMI			
Mean [SD]			
Min ; Max			
Smoking Status			
Never Smoked			
Current Smoker			
Ex-Smoker			
Does the patient have known diabetes?			
No			
Yes			
If yes, how is it managed?			
Diet-controlled			
Tablets			
Insulin			
Serum albumin level (g/L)			
Mean [SD]			
Min ; Max			
Is the patient on any immunosuppressive therapy?			
No			
Yes			
Is the patient clinically jaundiced? *(or serum bilirubin >50 µmol/L)*			
No			
Yes			
Does the patient have an active malignancy?			
No			
Yes			
Minimisation variables			
What degree of operative field contamination was found?			
Clean			
Clean-contaminated			
Contaminated			
Dirty			
Is a stoma present?			
No			
Yes			
If yes was it:			
Pre-existing			
Formed during this operation			
Surgery data			
ASA physical status clarification			
ASA I			
ASA II			
ASA III			
ASA IV			
ASA V			

Were prophylactic antibiotics given?			
No			
Yes – During procedure			
Yes – On induction			
Yes - On induction & During procedure			
Is the patient going to continue antibiotics post-operatively?			
No			
Yes			
Skin preparation used			
Aqueous betadine			
Alcoholic betadine			
0.5% Aqueous Chlorhexidine			
0.5% Alcoholic Chlorhexidine			
2% Aqueous Chlorhexidine			
2% Alcoholic Chlorhexidine			
Other			

COVID-19 data	**SUNPD (N=)**	**Control (N=)**	**Total (N=)**
Any COVID-19 symptoms on the day of surgery?			
No			
Yes			
Proven antibodies to SARS-CoV-2?			
No			
Yes			
SARS-CoV-2 virus status on the day of surgery?			
Screened positive			
Screened negative			
Screened but results unknown			
Not screened			
Positive SARS-CoV-2 swab result or clinical diagnosis of COVID-19?			
No or Unknown			
Yes			
If Yes, how long before surgery was the diagnosis			
Day of Surgery			
1-7 days			
8-14 days			
15-28 days			
5-6 weeks			
7-8 weeks			
3-4 months			
5-6 months			
6+ months			

89

Appendix D3: Description of the intervention(s)

Other Surgery Data	SUNPD (N=)	Control (N=)	Total (N=)
Actual procedure performed			
Adhesiolysis			
Appendicectomy			
Cholecystectomy			
Etc…			
Bowel – colonic			
Bowel – non-colonic			
Non-bowel			
N/A			
What was the surgical approach?			
Open (midline)			
Open (non-midline)			
Laparoscopic assisted/Laparoscopic converted			
Actual length of the incision (cm)			
Mean [SD]			
Min - Max			
<15 cm			
≥15 cm			
Was the WHO surgical safety checklist used?			
No			
Yes			
Does the patent have documented MRSA colonisation?			
No			
Yes			
Was malignancy present?			
No			
Yes/Suspected			
What was the estimated blood loss?			
<100ml			
100-500ml			
501-1000ml			
>1000ml			
Was an on-table blood transfusion required?			
No			
Yes			
Was the patient on inotropes at the end of the operation?			
No			
Yes			
Was a wound edge protection device used?			
No			
Yes			
Were triclosan impregnated sutured used?			
No			
Yes			
Were catheters left in place for local anaesthetic infiltration?			
No			
Yes			
Was adhesive or 'incise' drape used?			
No			
Yes (iodine-impregnated)			
Yes (plain incise drape)			

Was a wound/incision wash performed? No Yes (Betadine) Yes (Saline/Water) Yes (Other)			
Before closing, were gloves changed? No Yes			
Before closing, were instruments changed? No Yes			
How was the skin closed? Staples Interrupted sutures Continuous sutures			
Grade of operating surgeon? Consultant Registrar level SHO level ANP level			
Grade of surgeon closing fascia? Consultant Registrar level SHO level ANP level			
Grade of surgeon closing skin? Consultant Registrar level SHO level ANP level			
Total duration of operation (mins) Mean [SD] Min ; Max			

90

Appendix D4: Adherence to allocated intervention

Adherence to allocation intervention	SUNPD (N=)	Control (N=)	Total (N=)
What was the dressing applied to the laparotomy wound?			
SUNPD			
Conventional occlusive dressing			
Skin glue			
No dressing			
Other			
If SUNPD applied, what was the size of the dressing:			
10cm x 20cm			
10cm x 30cm			
10cm x 40cm			
Other			
Was the dressing applied in accordance with the randomised allocation?			
No			
Yes			

Reason for non-adherence in SUNPD arm (N=)

- Reason 1
- Reason 2
- Etc. …

Reason for non-adherence in control arm (N=)

- Reason 1
- Reason 2
- Etc. …

The per-protocol set:

- For participants randomised to the SUNPD arm, they will be deemed adherent to their randomised allocation if they received the SUNPD and the dressing was on for at least 3 days or until day of discharge if discharged earlier than 3 days.
- For the control arm, participants will be deemed adherent if they do not receive the SUNPD and receive the surgeon's preference of dressing (which may be conventional occlusive dressings, skin glue or no dressing but not another SUNPD).

Adherence to allocation intervention	SUNPD (N=)	Control (N=)	Total (N=)
Adherence as per the definition of the per-protocol set			
No			
Yes			
Unable to compute due to missing data for date of operation			
Unable to compute due to Day-7 (discharge) form missing			
Unable to compute due to missing data on date of final SUNPD removal			
N (%) of patient experiencing a skin reaction to the dressing?			
N (%) of patient experiencing any pain/discomfort related to the dressing?			

91

Appendix D5: Protocol deviations

	Allocated intervention	
Protocol deviation	**SUNPD (N=)**	**Control (N=)**
…		
…		
…		

Appendix D6: Primary outcome results

Primary outcome	SUNPD (N=)	Control (N=)	Total (N=)	Adjusted Relative Risk*			Adjusted Risk difference*		
				Effect size	95% CI	p-value	Effect size	95% CI	p-value
SSI within 30 days of surgery									
No									
Yes									

*Analysis adjusted for all minimisation variables with centre included as a random effect

Control arm is the reference category and so values less than 1 for the risk ratio and values less than 0 for the risk difference indicates better outcomes for SUNPD arm

<u>Breakdown of SSI reporting in SUNPD group</u>

- Number of participants with SSI reported at Day 7 (or discharge) –
- Number of participants with SSI reported at Return to Theatre –
- Number of participants with SSI reported at Day 30 –
- Number of participants with SSI reported from Patient Diary –
    - ***Number of participants with SSI reported from Patient Diary only –***

<u>Breakdown of SSI reporting in Control group</u>

- Number of participants with SSI reported at Day 7 (or discharge) –
- Number of participants with SSI reported at Return to Theatre –
- Number of participants with SSI reported at Day 30 –
- Number of participants with SSI reported from Patient Diary –
    - ***Number of participants with SSI reported from Patient Diary only –***

93

Appendix D7: Secondary outcomes results

Binary secondary outcomes	SUNPD (N=)	Control (N=)	Total (N=)	Adjusted Relative Risk*			Adjusted Risk difference*		
				Effect size	95% CI	p-value	Effect size	95% CI	p-value
Wound complications within 30 days									
No									
Yes									
Hospital re-admission for wound related complications within 30 days									
No									
Yes									

*Analysis adjusted for all minimisation variables with centre included as a random effect

Control arm is the reference category and so values less than 1 for the risk ratio and values less than 0 for the risk difference indicates better outcomes for SUNPD arm

Wound complications within 30 days of surgery as graded by the Clavien-Dindo scale

Wound complication	Number of patients with any wound complication by Clavien-Dindo grade				
	GRADE I *(None/conservative management or On ward intervention)*	GRADE II *(Antibiotic drug treatment)*	GRADE III *(Radiological or Surgical intervention)*	GRADE IV *(ITU admission)*	GRADE V *(Death)*
SUNPD arm					
Granuloma					
Seroma					
Haematoma					
Dehiscence					
Other…					
Control arm					
Granuloma					
Seroma					
Haematoma					
Dehiscence					
Other…					

Wound complications as graded by the Clavien-Dindo scale

Continuous data secondary outcomes	Statistic	Time point	SUNPD	Control	Total	Adjusted mean difference (95% CI) P-value
Length of hospital stay (UK patients only)	N Mean [SD] Min – Max	-				
SF-12 Physical component score (PCS)	N Mean [SD] Min – Max	Baseline				-
	N Mean [SD] Min – Max	Day 7				
	N Mean [SD] Min – Max	Day 30				
SF-12 Mental component score (MCS)	N Mean [SD] Min – Max	Baseline				-
	N Mean [SD] Min – Max	Day 7				
	N Mean [SD] Min – Max	Day 30				
Patient reported pain at the site of their primary laparotomy	N Mean [SD] Min – Max	Day 7				
	N Mean [SD] Min – Max	Day 30				

	EQ-5D-5L EuroQol score	N Mean [SD] Min – Max	Baseline				-	
		N Mean [SD] Min – Max	Day 7					
		N Mean [SD] Min – Max	Day 14					
		N Mean [SD] Min – Max	Day 21					
		N Mean [SD] Min – Max	Day 30					
	EQ-5D-5L Health score	N Mean [SD] Min – Max	Baseline				-	
		N Mean [SD] Min – Max	Day 7					
		N Mean [SD] Min – Max	Day 14					
		N Mean [SD] Min – Max	Day 21					
		N Mean [SD] Min – Max	Day 30					

Patient acceptability of use of their dressing	N Mean [SD] Min – Max	Day 7				
Health professional's acceptability of use of SUNPD (UK only)	N Mean [SD] Min – Max	-				

$-mean difference adjusted for all minimisation variables with centre as a random effect and baseline score (where available)

Appendix D8: Safety

Number of patients experiencing at least one SAE

Patients experiencing any SAE	SUNPD (N=)	Control (N=)	Total (N=)	Adjusted RR$^{\$}$ (95% CI) P-value
No Yes				

\$-relative risk adjusted for all minimisation variables
Reference group in the analysis is control arm so RR>1 values indicate worse outcome for SUNPD arm

Total number of SAE's

Total number of SAE's	SUNPD (N=)	Control (N=)	Total (N=)
Number of patients experiencing exactly: *1 SAE*			
2 SAE's			
… SAE's			
Total number of SAE's			

\$-incidence rate ratio adjusted for all minimisation variables
Reference group in the analysis is control arm so higher values indicate worse outcome for SUNPD arm

SAE listing by SAE category

SAE Category	SUNPD (N=)		Control (N=)		Total (N=)	
	Patients N (%)	Events N	Patients N (%)	Events N	Patients N (%)	Events N
Prolongation of hospitalisation						
Wound infection (SSI)						
Other complications (like Granuloma, Seroma, etc)						
Re-admission to hospitalisation						
Wound infection (SSI)						
Other complications (like Granuloma, Seroma, etc)						
Other expected SAE's						
Anastomotic leak						
Intra-abdominal sepsis						
Intra-peritoneal collection						
Thrombo-embolic event						
Infection not related to the wound						
Cardiac or central nervous complication						
All other SAE's						
SAE Name 1						
SAE Name …						
Total	-		-		-	

95

Appendix D9: Subgroup analysis for primary outcome

Subgroup analysis	SUNPD (N=)	Control (N=)	Interaction p-value	Risk Ratio (95% C.I)
Degree of contamination				
Clean				
Clean contaminated				
Contaminated				
Dirty				
Whether or not a stoma is present				
No				
Yes				
Operative procedure				
Bowel – colonic				
Bowel – non-colonic				
Non-bowel				
BMI				
Underweight (<18.5)				
Healthy weight (18.5-24.9)				
Overweight (25.0-29.9)				
Obese (≥30.0)				
Size of wound				
≤15cm				
>15cm				
Skin preparation used				
2% Alcoholic Chlorhexidine				
Aqueous betadine				
All other skin preparation				
Country				
UK				
Australia				
Day 30 assessment method				
Wound visualisation in-person				
Wound visualisation remotely either by video or photo				
Remotely without visualisation				

Subgroup analysis forest plot

96